

EcoDialysis — Executive Summary

ANDCS × MIT Hacking Medicine · Track 1 Sustainable Care · Paris 2026

The hidden water vulnerability of dialysis

Hook A 100-patient dialysis centre quietly discharges ~4,200 m³/yr of high-purity RO reject — water already paid for, treated, and lost. Décret n°2024-796 (in force 1 Sept 2024) now authorises non-clinical reuse via the EICH framework. France can turn dialysis from a hidden water risk into a measurable water-resilience asset.

~4,200 m³ RO reject / yr / centre	2–4 yr indicative payback	~30 k€ capex anchor (Saint-Exupéry)	1,128 m³/yr French measured precedent	156 sessions per patient / yr
What we deliver 4-point metrology install (feed, permeate, reject, reuse). - Tailored reuse map: green / orange / red uses, by site. - EICH dossier + ARS pre-instruction package. - Hardware integration (tank, UV-C, backflow, signage, sanitary log). - SaaS audit trail: KPIs, m³ saved, € verified, claims discipline.			Proof base - UK: Canterbury (800 L/h, £7.5k/yr), Bristol (13,140 m³/yr), Ashford design-in £2.5k. - ANZ: ANZSN cluster — >160,000 L & 6.4 MWh/yr saved. - Spain: 20-centre mean 476.5 L/session. - FR: Saint-Exupéry ~1,128 m³, ~30 k€, 5–6 yr. - SFNDT 2026: Qd 500 → 400 ml/min saves ~24 L dialysate + 60–100 L pretreated/session.	

Three first pilots — what we need

CHU / CHRU Reference flagship, full EICH path, public-sector credibility.	GHT private group Replicability across 6–10 centres, procurement framework.	Single large private centre Speed: 18-week timeline, retrofit + RO replacement coupling.
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Ask: 3 pilot LOIs · access to water bills, RO logs, session schedule · sponsor (direction or technical services) · ARS pre-instruction support.

Discipline — what we refuse to do

- No clinical-contact reuse. No used-dialysate recycling.
- No "carbon neutral" or "zero impact" claims. Water carbon caveat: ~1.0–1.1 tCO₂e on 4,200 m³ — real but secondary; main case is resilience & OPEX.
- No vendor lock-in. Neutral metrology, neutral integrator.

Sources: SFNDT 2023 & 2026 guides · Décret n°2024-796 / EICH · Planification écologique du système de santé · The Shift Project · IRDES · HCWH · deep-research syntheses (Perplexity / OpenAI / Gemini). Full diligence library in the appendix of the main deck (slides A1–A16 + source registry).